Genesis OS / Relationship Scores Readiness & Forecast Contract v1

Part 2 Appendix Pack

Engineering appendices freezing formulas, pair-state outputs, drift eligibility mechanics, and safe user-facing language.

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Authority note. This appendix pack binds to Readiness & Forecast Contract v1. When workbook math, approved scoring service output, and these appendices conflict, resolution order is: **approved workbook / scoring service** > **main contract** > **appendix pack** > **UI copy.** These appendices exist to remove gray area for engineering while preserving the workbook as the canonical scoring source where already frozen.

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Appendix A - Readiness Formula Appendix

A1. Purpose

This appendix defines the v1 engineering formula for Readiness, including fallback behavior when the system has only assessment data, confidence scoring, suppression rules, and the exact contract for how penalties are applied. The objective is consistent public-safe readiness scoring without pretending that sparse evidence is rich evidence.

A2. Canonical readiness question

Readiness measures how prepared a user is, at the current evidence state, to participate in a healthy relationship without forcing another person to carry unmanaged contradiction, instability, active shadow pressure, or unresolved pattern disruption.

A3. Stored fields

Field	Type	Range	Purpose
readiness_score	int	0-100	Canonical user-level output used for public surfaces once display policy allows.
readiness_confidence	int	0-100	Confidence attached to the readiness score.
readiness_sufficiency_status	enum	insufficient | internal_only | user_visible	Prevents premature exposure.
readiness_reason_codes	string[]	n/a	Machine-readable explanation for penalties or suppression.
readiness_version_id	string	n/a	Formula and policy version for reproducibility.
readiness_evidence_window	json	n/a	Counts, recency, and source families used in the score.

A4. Component families

All component inputs are normalized to 0.00-1.00 before weighting.

Family	Symbol	Weight	Primary inputs	Notes
Assessment foundation	AF	0.34	Question set outputs, archetype-appropriate high-function responses, values / intention alignment	Used even in assessment-only mode.
Consistency integrity	CI	0.18	Contradiction clusters, reflection completion, self-report symmetry, improbable self-presentation flags	Protects against polished-but-false self-description.
Behavioral regulation	BR	0.22	Messaging reciprocity, repair behavior, escalation control, response stability, observed activation patterns	Used only when behavior evidence exists.
Companion stability	CS	0.16	Drift state, grief state, anger state, ExsAnonymous state, emotional genome overlays	Companion systems can stabilize or reduce trust in the score.
Reflection and repair capacity	RR	0.10	Sanctuary reflections, accountability responses, post-	Small but meaningful weight; cannot rescue severe

			rupture repair participation	instability alone.

A5. Normalization rules

- Every component family must resolve to a scalar in the closed interval [0.00, 1.00].
- If a family has no valid evidence, it is marked unavailable rather than silently treated as zero.
- If only assessment evidence exists, use the assessment-only fallback formula below rather than pretending behavioral or companion evidence exists.
- If a family has stale evidence outside its recency window, confidence falls even if the last known family value remains cached for internal use.

A6. Primary formula

```
readiness_base_01 = (0.34 * AF) + (0.18 * CI) + (0.22 * BR) + (0.16 * CS) + (0.10 * RR)
readiness_penalty_01 = contradiction_penalty + shadow_penalty + drift_penalty +
acute_state_penalty
readiness_penalty_01 is capped at 0.35 in v1
readiness_final_01 = clamp(readiness_base_01 - readiness_penalty_01, 0.00, 1.00)
readiness_score = round(readiness_final_01 * 100)
```

A7. Assessment-only fallback

When the system has no reliable behavior stream yet, readiness may still be generated from onboarding data, but it must carry lower confidence and stricter display policy.

```
assessment_only_base_01 = (0.70 * AF) + (0.30 * CI)
assessment_only_penalty_01 = contradiction_penalty + declared_instability_penalty
assessment_only_penalty_01 is capped at 0.20
assessment_only_score = round(clamp(assessment_only_base_01 - assessment_only_penalty_01,
0.00, 1.00) * 100)
```

A8. Penalty contract

Penalty family	Symbol	Default range	Trigger	Cap rule
Contradiction penalty	CP	0.00-0.12	Active reflection clusters, unresolved answer inconsistencies, repeated self-report mismatch	Max 0.12
Shadow pressure penalty	SP	0.00-0.10	Any shadow_effective	Max 0.10

			_k meaningfully active, weighted by dominance and persistence	
Drift penalty	DP	0.00-0.08	Provisional or active drift pressure undermining identity stability or state trust	Max 0.08
Acute state penalty	AP	0.00-0.10	High active grief / anger / post-rupture acute disruption	Max 0.10

Penalty implementation notes:

- Penalties reduce readiness; they never increase it.
- A user cannot be penalty-free if a severe contradiction cluster is active and unresolved.
- Shadow penalties use shadow_effective_k, not one-event activation spikes.
- Companion modules may contribute to CS and may also impose an acute penalty when the state is currently disruptive.

A9. Example deterministic mappings

```
contradiction_penalty = min(0.12, 0.03 * active_cluster_count + 0.02 *
high_severity_cluster_count)
```

```
shadow_penalty = min(0.10, 0.06 * dominant_shadow_effective + 0.04 *
second_shadow_effective)
```

```
drift_penalty = 0.00 if drift_state in {stable, none}
```

```
drift_penalty = 0.03 if drift_state == provisional_watch
```

```
drift_penalty = 0.05 if drift_state == active_micro_drift
```

```
drift_penalty = 0.08 if drift_state == active_slow_drift or hold_unstable
```

A10. Confidence formula

Confidence is attached to readiness as a first-class value, not a hidden QA field.

```
confidence_01 = (0.30 * evidence_coverage) + (0.25 * recency_quality) + (0.20 * cross-
source_agreement) + (0.15 * behavior_depth) + (0.10 * stability_of_components)
```

```
readiness_confidence = round(clamp(confidence_01, 0.00, 1.00) * 100)
```

Confidence band	Numeric band	Policy
Low	0-49	Internal only. No public

		number.
Guarded	50-64	User may see a guarded band label, not raw precision.
Usable	65-79	User may see rounded readiness and safe explanatory copy.
Strong	80-100	User may see readiness plus deeper explanation or premium insight.

A11. Sufficiency gates

- user_visible readiness requires at least one complete assessment, contradiction processing, and either behavior evidence or a declared onboarding-only label.
- Any readiness score with confidence below 50 is internal_only, even if a number exists.
- If active drift is unresolved, readiness may remain visible only as a band label unless product policy explicitly authorizes numeric display.
- If no evidence has been refreshed within the configured recency window, the score is stale and must not be shown as current.

A12. Public bands

Score	Band	User-facing label	Meaning
0-39	Red	Not relationship-ready right now	High instability, contradiction, or active disruption.
40-54	Amber	Needs grounding before deeper connection	Some capacity is present, but reliable readiness is not yet stable.
55-69	Yellow-Green	Emerging readiness	Functional signs exist, but the system still sees pressure points.
70-84	Green	Relationship-ready with normal growth edges	Generally healthy operating range.

85-100	Deep Green	Strong readiness signal	High-functioning current profile with solid evidence support.

A13. Engineering note

If the workbook already freezes a more granular field-level formula, engineering must use that formula and map its output into this contract's storage, confidence, sufficiency, and display rules. This appendix is intended to stop ambiguous implementations, not overrule a more specific canonical scoring sheet.

Appendix B - Forecast Output Appendix

B1. Forecast is a governed output family

Forecast does not mean destiny. In v1 it is a bundle of pair-state outputs derived from pair evidence, match structure, repair behavior, pacing, shadow pressure, drift pressure, and confidence. The system may route product decisions from Forecast, but must never describe it as certainty.

B2. Required forecast outputs

Output	Field	Range	Refresh trigger	Visible to users?
Pair stability outlook	forecast_pair_stability	0-100	New pair event, compatibility refresh, drift state change	Yes, if confidence allows
Rupture risk 30d	forecast_rupture_risk_30d	0-100	Tension / escalation / repair events	Usually banded
Repair capacity	forecast_repair_capacity	0-100	Repair attempts, apology behavior, post-rupture actions	Yes, if confidence allows
Reciprocity health	forecast_reciprocity_health	0-100	Messaging / response / effort balance updates	Yes, if confidence allows
Pacing alignment	forecast_pacing_alignment	0-100	Intentions, cadence, mismatch events, ask / avoid patterns	Yes, if confidence allows
Shadow pressure index	forecast_shadow_pressure	0-100	Shadow activation / persistence refresh	Internal or soft-explained
Drift pressure index	forecast_drift_pressure	0-100	Drift state changes	Mostly internal / soft-explained

Forecast confidence	forecast_confidence	0-100	Any forecast recompute	Always required, not always shown

B3. Normalized source families

```
compat_norm = clamp((compatibility_score - 350) / 500, 0.00, 1.00)
```

```
pair_rhythm_norm = normalized pair rhythm from cadence, reciprocity, and mutual continuation patterns
```

```
repair_norm = normalized repair capacity from apology quality, follow-through, and post-conflict recovery
```

```
conflict_norm = normalized communication / conflict alignment
```

```
shadow_load_norm = normalized pair shadow pressure from both users' shadow_effective values and event activations
```

```
drift_load_norm = normalized pair drift pressure from both users' drift states and eligibility posture
```

```
acute_pressure_norm = normalized acute disruption pressure from grief / anger / rupture recency
```

B4. Pair stability outlook formula

```
pair_stability_base_01 = (0.40 * compat_norm) + (0.20 * pair_rhythm_norm) + (0.20 * repair_norm) + (0.10 * conflict_norm) + (0.10 * reciprocity_norm)
```

```
pair_stability_penalty_01 = (0.30 * shadow_load_norm) + (0.25 * drift_load_norm) + (0.20 * acute_pressure_norm) + (0.25 * contradiction_load_norm)
```

```
forecast_pair_stability = round(clamp(pair_stability_base_01 - pair_stability_penalty_01, 0.00, 1.00) * 100)
```

B5. Rupture risk 30d formula

```
rupture_risk_base_01 = (0.35 * escalation_rate_30d) + (0.20 * unresolved_tension_density) + (0.15 * low_repair_rate) + (0.15 * shadow_load_norm) + (0.15 * drift_load_norm)
```

```
forecast_rupture_risk_30d = round(clamp(rupture_risk_base_01, 0.00, 1.00) * 100)
```

B6. Repair capacity formula

```
repair_capacity_01 = (0.40 * repair_success_rate) + (0.20 * apology_follow_through) + (0.15 * reduced_latency_norm) + (0.15 * mutual_repair_participation) + (0.10 * reflection_completion_pair)
```

```
forecast_repair_capacity = round(clamp(repair_capacity_01, 0.00, 1.00) * 100)
```

B7. Reciprocity and pacing formulas

```
reciprocity_health_01 = (0.45 * mutual_initiation_balance) + (0.25 * response_reciprocity) + (0.15 * effort_balance) + (0.15 * conversation_continuity)
```

```
pacing_alignment_01 = (0.40 * intention_alignment) + (0.25 * cadence_alignment) + (0.20 * exclusivity_or_depth_alignment) + (0.15 * pressure_mismatch_inverse)
```

```
forecast_reciprocity_health = round(clamp(reciprocity_health_01, 0.00, 1.00) * 100)
```

```
forecast_pacing_alignment = round(clamp(pacing_alignment_01, 0.00, 1.00) * 100)
```

B8. Confidence and visibility

```
forecast_confidence_01 = (0.35 * pair_event_depth) + (0.20 * recency_quality) + (0.20 *
cross-signal_agreement) + (0.15 * duration_coverage) + (0.10 * evidence_diversity)
```

```
forecast_confidence = round(clamp(forecast_confidence_01, 0.00, 1.00) * 100)
```

Confidence	Display posture	User view	Internal use
0-49	Suppressed	No forecast number. Use 'not enough shared evidence yet.'	Allowed for internal QA only.
50-64	Guarded	Only soft labels and guidance, no raw risk number.	Allowed for routing and interventions.
65-79	Visible	Rounded outputs and short explanation.	Allowed.
80-100	Expanded	Rounded outputs plus deeper relational insight.	Allowed.

B9. Output thresholds

Metric	Strong	Guarded	Risk zone
Pair stability outlook	70-100	55-69	0-54
Repair capacity	70-100	55-69	0-54
Reciprocity health	70-100	55-69	0-54
Pacing alignment	70-100	55-69	0-54
Rupture risk 30d	0-29	30-49	50-100

B10. Suppression rules

- No user-visible Forecast when pair_event_depth is below the configured minimum or when forecast_confidence < 50.
- No raw rupture risk number on first-contact or low-evidence surfaces; use qualitative copy instead.
- Forecast is downgraded to guarded mode if either user is in hold_unstable drift state, acute grief / anger state, or high active contradiction state.
- Forecast cannot overrule explicit policy safety blocks or moderation actions.

B11. Relationship to match score

Compatibility and Forecast are siblings, not duplicates. Compatibility reflects structural fit. Forecast reflects current projected pair-state under recent evidence. A high compatibility score with poor recent repair behavior can still produce guarded or risky Forecast outputs.

Appendix C - DriftEligibility Technical Appendix

C1. Purpose

DriftEligibility governs when slow-changing identity-adjacent traits are allowed to move. Its purpose is to stop identity chaos, prevent one bad week from redefining a person, and distinguish short-term activation from durable pattern change.

C2. v1 stance

- Slow drift is OFF by default except for shadow_trait_prior micro-drift and attachment_stability_index.
- Archetype drift remains OFF in v1 production.
- Drift requires stable evidence and never fires in the middle of a storm.

C3. Inputs

```
rollup_n50: last 50 eligible events
rollup_n20: last 20 eligible events
rollup_prev50: previous 50-event window, if available
current_genome: current slow trait values and priors
config thresholds: environment-owned values with version tag
```

C4. Required rollup metrics

Metric	Symbol	Window	Definition	Threshold
Escalation rate	E50 / E20	50 / 20	Escalation events divided by eligible events in window	E20 <= 0.35
Repair success rate	R50	50	repair_event / tension_events	No hard gate; used in context and confidence
Repair latency normalized	L50	50	Mean normalized delay from rupture to repair	No hard gate; informs confidence and forecast
Arousal slope	A_slope50	50	Linear trend of arousal over window	abs(A_slope50) <= 0.05
Arousal volatility	A_std50	50	Standard	A_std50 <= 0.22

			deviation of arousal	
Contradiction severity mean	C50	50	Mean contradiction severity over window	C50 <= 0.55 OR improved by >= 0.10
Move entropy normalized	H_norm	50	Entropy of move distribution across withdraw / press / escalate / appease / repair / joke	H_norm <= 0.78
Event count	N50	50	Total eligible events available	N50 >= 50
Unique day spread	D50	50	Number of distinct days covered by rollup	D50 >= 10

C5. Move entropy formula

```
Let p_i be the normalized frequency of move categories in the last 50 events: withdraw,
press, escalate, appease, repair, joke.
```

```
H = - Σ p_i * ln(p_i)
```

```
H_norm = H / ln(K), where K is the number of move categories present
```

```
Lower entropy means the behavior pattern is more stable and therefore more eligible for
slow drift evaluation.
```

C6. Gate logic

- G0 Evidence minimum: N50 >= 50 and D50 >= 10.
- G1 Not currently unstable: E20 <= 0.35 and A_std50 <= 0.22.
- G2 Trend is settled: abs(A_slope50) <= 0.05.
- G3 Contradictions are not spiking: C50 <= 0.55 OR C50 improved by at least 0.10 versus previous 50-window.
- G4 Behavior distribution is not chaotic: H_norm <= 0.78.
- All gates must pass before slow drift is eligible.

C7. Eligibility confidence formula

```
conf = 0.25 * (1 - clamp(E20 / 0.35, 0, 1))
     + 0.20 * (1 - clamp(A_std50 / 0.22, 0, 1))
```

```
    + 0.20 * (1 - clamp(abs(A_slope50) / 0.05, 0, 1))
    + 0.20 * (1 - clamp(C50 / 0.55, 0, 1))
    + 0.15 * (1 - clamp(H_norm / 0.78, 0, 1))
eligible_final = gates_passed AND conf >= 0.60
```

C8. Reason codes

Reason code	Meaning
insufficient_events	Not enough events to support slow drift evaluation.
insufficient_day_spread	Events are too concentrated in time; one short period cannot redefine identity.
too_unstable_recently	Recent escalation rate is too high for safe drift reclassification.
high_volatility	Arousal volatility is too high; postpone slow drift.
trend_not_settled	Arousal slope suggests the user is still moving sharply rather than stabilizing.
contradictions_spiking	Contradictions remain too elevated or have not improved enough.
behavior_inconsistent	Move entropy indicates unstable or chaotic behavior distribution.

C9. Drift state machine

State	Meaning	Enter when	Exit when
none	No drift evidence yet	New user or no drift-eligible rollups	Evidence arrives
provisional_watch	Soft movement pressure exists but gates do not fully pass	Micro-drift pressure detected but slow drift gates not met	Pressure clears or gates pass consistently

hold_unstable	System is explicitly preventing drift because recent instability is too high	E20 > 0.35 or A_std50 > 0.22	Stability gates pass for two consecutive recomputes
eligible_micro_drift	Micro-drift may update shadow_trait_prior or attachment_stability_index with caps	Pressure exists and micro-drift rules pass	Pressure clears or stronger drift state applies
eligible_slow_drift	Slow drift may update allowed slow traits	All gates pass and conf >= 0.60	Stability fails or cooldown begins after apply
active_slow_drift	A permitted slow trait update has been applied during current cycle	eligible_slow_drift and capped update committed	Cooldown ends
cooldown	Prevent repeated reclassification thrashing	Any slow-drift apply event	Cooldown duration expires and new evidence accumulates

C10. Transition rules

- none -> provisional_watch when repeated micro-pressure is detected but evidence is still shallow.
- provisional_watch -> hold_unstable when E20 or A_std50 crosses instability gates.
- provisional_watch -> eligible_micro_drift when micro-drift-specific thresholds are met for at least two consecutive recomputes.
- eligible_micro_drift -> eligible_slow_drift only if all slow gates pass and conf >= 0.60.
- eligible_slow_drift -> active_slow_drift only after an update commit writes old value, delta, reason codes, and version id to the audit ledger.
- active_slow_drift -> cooldown immediately after apply. Cooldown prevents churn.
- cooldown -> provisional_watch or none once the cooldown duration expires and fresh evidence exists.

C11. Update caps

```
shadow_activation_recent_k = shadow_activation_recent_k + B * (a_k -
shadow_activation_recent_k), where B = 0.06
```

```
daily increase cap per shadow activation recent = +0.12
```

```
shadow_effective_k = clip(0.70 * shadow_trait_prior_k + 0.30 * shadow_activation_recent_k,
0.00, 1.00)
```

- One bad fight must not permanently brand a user.
- Meaningful activation requires persistence: treat a shadow as active only if a_k >= 0.70 in at least 3 of the last 10 eligible events.
- Slow trait updates should be capped per cycle and logged with before/after values.

C12. Pseudocode

```
function DriftEligibility(roll50, roll20, rollPrev50, currentState):
    reasons = []
    if roll50.total_events < 50: reasons += ['insufficient_events']
    if roll50.unique_days < 10: reasons += ['insufficient_day_spread']
    if roll20.escalation_rate > 0.35: reasons += ['too_unstable_recently']
    if roll50.arousal_std > 0.22: reasons += ['high_volatility']
    if abs(roll50.arousal_slope) > 0.05: reasons += ['trend_not_settled']
    contradiction_ok = (roll50.contradiction_mean <= 0.55) or ((rollPrev50 exists) and
((rollPrev50.contradiction_mean - roll50.contradiction_mean) >= 0.10))
    if not contradiction_ok: reasons += ['contradictions_spiking']
    if roll50.move_entropy_norm > 0.78: reasons += ['behavior_inconsistent']
    conf = compute_conf(...)
    eligible = (len(reasons) == 0) and (conf >= 0.60)
    return { eligible, reason_codes: reasons, confidence: conf }
```

Appendix D - User-Facing Copy Library

D1. Purpose

This library standardizes how readiness and forecast surfaces explain scores without sounding clinical, deterministic, or manipulative. Copy must be calm, specific, and non-prophetic. The product is allowed to give signals and guidance. It is not allowed to cosplay as fate.

D2. Copy rules

- Prefer signal, pattern, trend, current outlook, current evidence.
- Avoid destiny, guaranteed outcome, diagnosis, final truth, doomed, soulmate certainty.
- Use the present tense. Forecast speaks about current projected pattern, not permanent identity.
- Do not tell users another person is bad, broken, or unsafe as a medical-style verdict.
- When confidence is low, say so plainly and avoid numeric theater.

D3. Readiness copy library

Use case	Recommended copy	Do not use
Primary score card	Your current Readiness score is {score}. This is a signal of how steady and relationship-ready your current patterns look right now.	You are officially ready for love forever.
Band-only low confidence	We're still gathering enough real evidence to turn this into a stronger signal. For now, your readiness looks {band_label}.	This is your exact truth.
Readiness explanation	This score reflects your current mix of consistency, emotional regulation, and follow-through - not your worth.	This proves who you are.
Improvement nudge	Your score can move when your patterns move. Reflection, repair, and steadier behavior matter more than perfect answers.	Just retake the quiz until the number is higher.

Suppressed due to stale evidence	Your readiness signal is paused until we have fresher evidence.	Your readiness has disappeared.

D4. Forecast copy library

Use case	Recommended copy	Do not use
Forecast intro	Forecast shows the current outlook of this connection based on shared evidence, not a guaranteed future.	This relationship will definitely work.
Low evidence pair	Not enough shared evidence yet to show a full forecast. As this connection develops, the signal will sharpen.	We already know where this is going.
Strong pair stability	Current evidence suggests a strong stability outlook if both people keep showing up the same way.	You two are soulmates.
Elevated rupture risk	The current pattern shows pressure around conflict and recovery. That doesn't equal failure, but it does call for care.	This relationship is doomed.
Good repair capacity	You both show signs of coming back together after tension instead of getting stuck in it.	You will always work things out.

D5. Safe short labels

```
Readiness labels: not ready right now | needs grounding | emerging readiness |
relationship-ready | strong readiness signal
```

```
Forecast labels: early signal only | guarded outlook | mixed outlook | stable outlook |
strong outlook
```

```
Risk labels: low current risk | moderate pressure | elevated pressure | high pressure
```

D6. Sanctuary and coaching prompts

Scenario	Approved copy
Readiness dip	Your score dipped because the system sees

	more pressure than steadiness right now. That is a signal to slow down and reset, not a verdict about your future.
Contradiction cluster	Some of your answers and behaviors are pulling in different directions. Before your score can rise, the system needs to see those signals settle.
Forecast guarded	This connection still has more questions than proof. Move slower. Let behavior answer what chemistry can't.
Repair nudge	The strongest improvement here won't come from saying the right thing once - it will come from what happens after tension.

D7. Final rule

Any new surface that explains readiness or forecast must use this library or a reviewed descendant of it. Engineers may not improvise emotionally loaded copy directly in templates, notifications, or partner payloads.